VOL. I · NO. 4 MONDAYS & THURSDAYS

THE LAUNCH REPORT

Rocket launches, reentries and the business of spaceflight, worldwide

THURSDAY, OCTOBER 22, 2026 ALL TIMES EASTERN DAYLIGHT TIME (EDT)

THE WEEK IN LAUNCH

MMX is on its way to Mars; three launches, three successes since Monday

Japan's H3 sent the Martian Moons eXploration (MMX) probe toward Mars at 3:41 p.m. EDT Monday, a Falcon 9 flew the Bandwagon-5 rideshare from Florida on Tuesday morning, and Galactic Energy launched a Ceres-1S from the sea off eastern China that night. That brings 2026 to 246 orbital launch attempts, 239 of them successful, counting GCAT through October 15 and Launch Library 2 since.

Two launches stood down. Thunderstorms scrubbed Starlink 15-15 at Vandenberg on Wednesday, a day after a technical issue stopped it, and Rocket Lab postponed an Electron for HawkEye 360 to review "out-of-family telemetry data." Both have new dates this week, as do three Long March rockets.

I · RECENT LAUNCHES & RETURNS

H3 flies MMX on time, to plan

The Phobos sample-return probe separated 71 minutes after liftoff and is due at Mars in August 2027.



JAXA's HTV-X1 cargo spacecraft approaches the International Space Station on Oct. 29, 2025, four days after launching from Tanegashima on an H3-24, the same H3 version that carried MMX.
PHOTO: NASA · PUBLIC DOMAIN

H3-24: MMX

The H3 lifted off from Tanegashima Space Center at 3:41 p.m. EDT Monday (19:41 UTC). The second stage and MMX reached a parking orbit 10 minutes 40 seconds after liftoff, and the 4,480-kilogram spacecraft separated at 1 hour 11 minutes 12 seconds, on a 10-month trajectory that reaches Mars in August 2027, NASASpaceflight reported. The separation time matches the one in the Japan Aerospace Exploration Agency's (JAXA) launch plan. JAXA announced at 1:10 a.m. EDT Wednesday that MMX had completed its initial critical operations phase.

MMX carries 13 science instruments, among them NASA's MEGANE gamma-ray and neutron spectrometer and France's MIRS infrared spectrometer, and will release IDEFIX, a French-German rover, onto Phobos (NASASpaceflight). The P-Sampler mounted on one leg of the MMX lander will collect surface material with high-pressure nitrogen gas, Blue Origin, which supplies it, posted.

Falcon 9: Bandwagon-5



A Falcon 9 lifts off from Space Launch Complex 40 at Cape Canaveral with Crew-12 on Feb. 13, 2026. Bandwagon-5 flew from the same pad on Tuesday.
PHOTO: NASA/CORY S. HUSTON · PUBLIC DOMAIN

Falcon 9 lifted off from Space Launch Complex 40 at Cape Canaveral at 7:04 a.m. EDT Tuesday (Launch Library 2) with 32 rideshare payloads, a day late because of recovery-zone weather. Booster B1085 landed on A Shortfall of Gravitas on its 19th flight. The stack carried SpaceX's second Starfall reentry capsule, 3.1 meters wide and able to carry 1,000 kilograms, plus four Xona Space Systems navigation satellites, Impulse

Space's LEO Express 4 space tug, Planet's Pelican-12 imager and radar satellites from ICEYE, Capella Space and Umbra, NASASpaceflight reported. Rocket Lab said Wednesday that the Maxwell spacecraft it built for Viasat, which launched from Cape Canaveral on Tuesday, is in orbit (post).

Ceres-1S: Xiguang-4 01

Galactic Energy's Ceres-1S launched at 11:39 p.m. EDT Tuesday from the sea off Rizhao, Shandong, carrying Xiguang-4 01, a 0.5-meter-resolution Earth observation satellite for the Guangzhou government and Guangzhou University (Launch Library 2). It was the first mission run from a new local command center rather than the Taiyuan launch center, and Galactic Energy has now had 23 successes in 26 launches, NASASpaceflight reported.

II · LAUNCH SCHEDULE

Starlink tonight, three Long Marches by Saturday

Every orbital launch with a date between now and November 5, plus missions targeting later this month. Times are EDT, with Coordinated Universal Time (UTC) below; NET means no earlier than.

DATE / TIME	VEHICLE	PROVIDER	MISSION / PAYLOAD	SITE	ORBIT	STATUS	SOURCES
Thu Oct 22, window opens 5:45 p.m. 21:45 UTC	Falcon 9	SpaceX	Starlink 15-15: 27 Starlink V2 Mini satellites. Booster B1063, 36th flight	SLC-4E, Vandenberg, Calif.	LEO	CONFIRMED	FAA; Spaceflight Now
Fri Oct 23, 3:00 a.m. 07:00 UTC	Electron	Rocket Lab	"We're Not Hawking Around": three HawkEye 360 satellites, 98th Electron	Launch Complex 1, Mahia, New Zealand	Sun-synchronous	TENTATIVE	Launch Library 2
Fri Oct 23, 6:15 a.m. 10:15 UTC	Long March 6A	CASC	Payload not announced	LC-9A, Taiyuan, China	Not announced	TENTATIVE	Launch Library 2
Fri Oct 23, 3:25 p.m. 19:25 UTC	Long March 4B	CASC	Payload not announced	Launch Area 94, Jiuquan, China	Not announced	TENTATIVE	Launch Library 2
Sat Oct 24, 9:50 p.m. 01:50 UTC Oct 25	Long March 8A	CASC	Payload not announced	Commercial LC-1, Hainan, China	Not announced	TENTATIVE	Launch Library 2
Mon Oct 26, 1:00 a.m. 05:00 UTC	Falcon 9	SpaceX	Starlink 15-16: 27 Starlink V2 Mini satellites	SLC-4E, Vandenberg, Calif.	LEO	TENTATIVE	FAA; Launch Library 2
NET Thu Oct 29	Vulcan VC6L	United Launch Alliance (ULA)	Amazon Leo LV-01 broadband satellites	SLC-41, Cape Canaveral, Fla.	LEO	PRELIMINARY	Launch Library 2
Thu Oct 29, 12:17 a.m. 04:17 UTC	Falcon 9	SpaceX	Starlink 15-18: 27 Starlink V2 Mini satellites	SLC-4E, Vandenberg, Calif.	LEO	TENTATIVE	Launch Library 2
NET October	GSLV Mk II	Indian Space Research Organisation (ISRO)	NVS-03 (IRNSS-1L) navigation satellite	Second Launch Pad, Sriharikota, India	GTO	PRELIMINARY	Launch Library 2
NET October	SSLV	ISRO	AayulSAT-1A and others	First Launch Pad, Sriharikota, India	LEO	PRELIMINARY	Launch Library 2
NET Nov 3	Long March 12A	CASC	Flight 2, with a first-stage landing attempt	Jiuquan, China	LEO	PRELIMINARY	NASASpaceflight; Launch Library 2 (October)
NET Wed Nov 4	Electron	Rocket Lab	"The Sea God Surveys": iQPS radar satellite	Launch Complex 1, Mahia, New Zealand	LEO	PRELIMINARY	Launch Library 2
Thu Nov 5, 5:34 p.m. EST 22:34 UTC	Falcon 9	SpaceX	NROL-195, National Reconnaissance Office	SLC-4E, Vandenberg, Calif.	LEO	TENTATIVE	Launch Library 2

CONFIRMED date and time announced by the launch provider, its customer or the FAA. **TENTATIVE** specific date in published schedules or airspace and maritime notices, not yet confirmed by the operator. **PRELIMINARY** target range, or no-earlier-than date or month only.

Abbreviations: SLC, Space Launch Complex; SLC-4E, Space Launch Complex 4 East; LC, Launch Complex; LEO, low Earth orbit; GTO, geostationary transfer orbit; GSLV, Geosynchronous Satellite Launch Vehicle; SSLV, Small Satellite Launch Vehicle; CASC, China Aerospace Science and Technology Corp.; EST, Eastern Standard Time, which begins November 1.

Date changes

Starlink 15-15. Tuesday, October 20, at 6:14 p.m.; scrubbed that evening for an apparent technical issue (Spaceflight Now), then on Wednesday at 7:04 p.m.: "With thunderstorms approaching the launchpad, SpaceX is scrubbing the launch of Starlink 15-15 for today" (Spaceflight Now). Now Thursday, window 5:45 to 10:28 p.m. (FAA).

Electron for HawkEye 360. NET Wednesday, October 21; Rocket Lab stood down at 12:55 a.m. EDT "to review out-of-family telemetry data observed during vertical operations on the pad" (Rocket Lab). Now Friday at 3:00 a.m., to be confirmed (Launch Library 2).

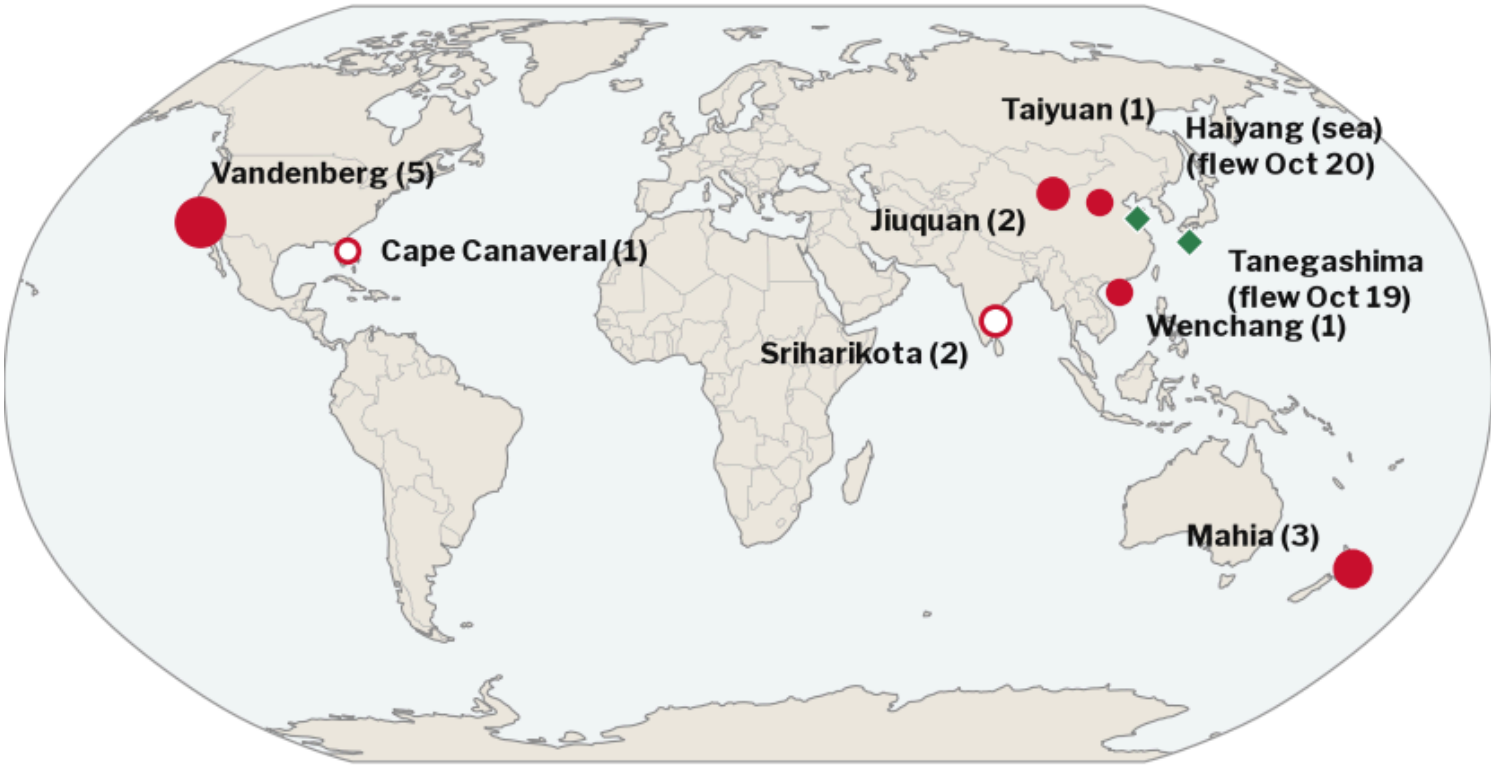
Long March 8A. Wednesday, October 21, at 10:57 p.m.; now Saturday, October 24, at 9:50 p.m. No reason given (Launch Library 2).

Starlink 15-16. Saturday, October 24, at 1:29 a.m.; now Monday, October 26, at 1:00 a.m. No reason given (FAA).

Long March 12A. Listed for October; NASASpaceflight reports NET November 3.

QUICK HIT

Starship. Flight 15 remains NET November 8, according to NASASpaceflight. Pad 2's booster quick disconnects were purging on Wednesday, "hopefully ahead of Booster 22 arriving soon," NASASpaceflight posted.



Launch sites with orbital launches listed through November 15; numbers count launches. Filled: dated launch (Confirmed or Tentative). Hollow: Preliminary. Green diamonds: sites that launched since Monday with nothing scheduled; Cape Canaveral also launched Bandwagon-5 on Tuesday. Haiyang is the port for Galactic Energy's sea launches off Rizhao.

MAP: THE LAUNCH REPORT · PAD COORDINATES FROM LAUNCH LIBRARY 2 · NATURAL EARTH

III · TRAJECTORIES & HAZARD AREAS

Airspace windows for two Starlinks and a Mojave flight

The FAA's operations plan lists the times U.S. airspace is set aside; no hazard-area coordinates have been published for these flights.

How MMX flew

H3 Flight 10, the H3-24L version, followed the eastward ascent in JAXA's plan, using two LE-9 hydrogen engines, four solid boosters and the long fairing. MMX separated at 1 hour 11 minutes 12 seconds, at 1:11:12 in the plan's timeline, after a parking-orbit insertion at 10 minutes 40 seconds, NASASpaceflight reported. The plan's drop zones for the boosters, fairing and first stage were mapped in Monday's issue.

Bandwagon-5 followed a southeasterly track from Cape Canaveral, with payload deployment beginning 57 minutes 18

seconds after liftoff and the Starfall capsule scheduled to separate last, at 2 hours 48 minutes 28 seconds (NASASpaceflight). Initial Starfall tests are planned to splash down in the Pacific about 1,300 kilometers off the United States or Mexico.

Airspace windows

The FAA Air Traffic Control System Command Center's plan for Thursday lists these launch windows, converted from UTC (FAA advisory):

FLIGHT	PRIMARY WINDOW	BACKUP
Starlink 15-15, Vandenberg	**Thu Oct 22, 5:45–10:28 p.m.**	**Fri Oct 23, 5:39–10:13 p.m.**
Starlink 15-16, Vandenberg	Mon Oct 26, 1:00–5:43 a.m.	Tue Oct 27, 12:45–5:28 a.m.
Stratolaunch "Erebus," Mojave Air and Space Port	Mon Oct 26, 9:00 a.m.–1:00 p.m.	Mon Nov 2, 9:00 a.m.–1:00 p.m. EST

IV · REENTRIES & ISS TRAFFIC

Cosmos 2572 is down; Rassvet-3 17 is now due next week

Three objects reentered on October 19 and 20. An Electron kick stage and a Chinese satellite are next.

Confirmed reentries

CelesTrak's satellite catalog lists these objects as decayed since Monday's issue; times are from Aerospace.

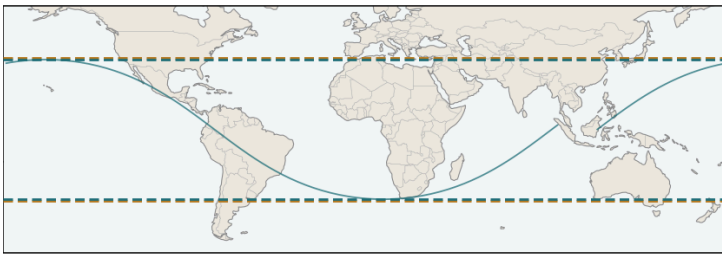
OBJECT (CATALOG NO.)	LAUNCHED	REENTRY, EDT	LAST ORBIT, KM
Cosmos 2572 (58435)	**Nov 25, 2023**	**Oct 19, 2:34 a.m. ± 1 hr**	**119 × 131**
Starlink-2610 (48376)	May 4, 2021	Oct 19, 1:13 p.m. ± 1 hr	156 × 158
Starlink-1305 (45377)	Mar 18, 2020	Oct 19, 11:23 p.m. ± 2 hr (predicted)	172 × 183

Five more older Starlink satellites had predicted reentries on October 20 and 21, but CelesTrak had not yet marked them decayed (Aerospace).

Predicted

OBJECT (CATALOG NO.)	LAUNCHED	PREDICTED, EDT
Electron kick stage (100684)	Sep 11, 2026	Fri Oct 23, 4:13 a.m. ± 16 hr
Yaogan-35 02B (52908)	Jun 23, 2022	Sat Oct 24, 7:38 a.m. ± 20 hr
Rassvet-3 17 (100083)	**Jul 19, 2026**	**Fri Oct 30, 6:56 a.m. ± 49 hr**

The Rassvet-3 17 broadband satellite, which Monday's issue listed for about noon Thursday, is now predicted for October 30 (Aerospace). The Electron kick stage is from Rocket Lab's September 11 "Happily Ever Faster" mission.



— Electron kick stage 36.0° — Yaogan-35 02B 35.0° — Rassvet-3 17 82.3°

Latitude bands within which each predicted object can reenter, set by its orbital inclination, with one orbit of Yaogan-35 02B computed from CelesTrak's elements from 4:51 a.m. EDT October 21. No current elements were available for the Electron kick stage or Rassvet-3 17.

MAP: THE LAUNCH REPORT · CELESTRAK · NATURAL EARTH

Station traffic

The CRS-35 cargo Dragon remains docked to the International Space Station, where it is to stay about a month (NASA). The next Russian cargo ship, Progress MS-36, is listed for 12:03 a.m. EST on November 24 from Baikonur, and Northrop Grumman's Cygnus NG-25 is NET November 30 (Launch Library 2).

V · INDUSTRY & REGULATION

China prepares its first booster reflight

A recovered Long March 10B stage is being readied to fly again, and Long March 12A will try another landing.

China

The Long March 10B first stage that was caught at sea by a net aboard the recovery ship Linghangzhe on July 10 is being refurbished in Tianjin and is expected to become China's first orbital-class booster to be reused. One of two new commercial pads in Hainan is to make its first launch, a Long March 10B, before the end of the year, and that flight is expected, though not officially confirmed, to use the recovered stage, NASASpaceflight reported. The second Long March 12A, NET November 3, will try to land its first stage after the first attempt crashed about five kilometers from its target when one of three engines did not restart.

CAS Space may fly its second three-core Lijian-2 before the end of October with 18 Qianfan broadband satellites, and a second Qingzhou cargo craft is to launch on Lijian-2 to the Tiangong station no earlier than January 2027 with up to 1,800 kilograms of supplies. Chinese orbital launches are up nearly 14% from this point last year (NASASpaceflight).

More Chinese launches are moving to sea. CAS Space is preparing the first seaborne Lijian-1, from the ship Dongfang Hangtiangang, which is to sail south for a launch off Guangdong in the South China Sea. Beijing-based i-Space expects to static-fire the first stage of its Shuang Quxian-3 before the end of the year and plans a droneship landing attempt on the rocket's first flight in early 2027, while Cosmoleap has been testing a Starship-style catch tower for its Yueqian-1 (NASASpaceflight).

Both Chinese broadband constellations are nearing service. The China Satellite Network Group's Guowang network reached 222 satellites with an October 9 launch and is trialling service, and Shanghai Spacesail plans to start Qianfan service in China by the end of the year (NASASpaceflight).

Rocket Lab and Firefly

Rocket Lab has rotated the Neutron second-stage tank assembly horizontal, "a critical step in the handling of large rocket sections for integration and assembly," and is building a new carbon-composite forward dome for the first-stage tank (post). Firefly Aerospace has completed the preliminary design review for its part in NASA's MoonFall mission, in which its Elytra spacecraft will deploy drones built by NASA's Jet Propulsion Laboratory above the Moon's south pole (post).

QUICK HIT

Blue and Gold. Next month the twin Mars spacecraft Rocket Lab built for NASA and the University of California, Berkeley, will swing past Earth and each fire its main engine in a single, continuous burn, using Earth's gravity, the company posted.

Europe

The 350th ESA Council appointed Andrzej Banasiak as Director of Commercialisation and Industry Partnership and Nicolaus Hanowski as Director of Space Operations, both from January 1, 2027, the European Space Agency posted Thursday. ESA is also developing OASIS, a lunar "service station" to provide power, communications and data to experiments through the two-week lunar night (post).

Comment deadlines

New Glenn at SLC-36. The FAA's virtual public meeting on raising New Glenn's cap from 12 to 50 launches a year is tonight from 6 to 8 p.m. EDT; comments close October 30 (notice, docket FAA-2026-9841).

Part 450. Comments on five proposed changes to the commercial launch and reentry licensing rule are due November 4 (docket FAA-2026-9935).

Starbase. Scoping comments on up to 456 Starship launches a year at Starbase are due November 30, with meetings November 17 to 19 (Federal Register). No new FAA launch notices have appeared since Monday (Federal Register).

VI · BY THE NUMBERS

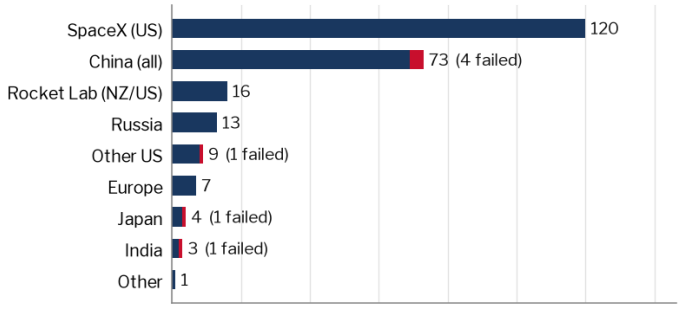
246 orbital launch attempts so far in 2026

All 13 orbital launches in October have succeeded. China needs 27 more to reach 100 in a year for the first time.

246	239	120	73	13
orbital launch attempts worldwide, Jan 1–Oct 21	reached orbit	by SpaceX, all successful	by China, four failures	orbital launches in October, all successful

The count

Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) records 243 orbital launch attempts through USSF-481 on October 15; the three launches since, logged by Launch Library 2, bring the year to 246, with 239 successes. SpaceX has flown 120, all successful, followed by China with 73, Rocket Lab with 16 and Russia with 13. Japan's MMX launch was its fourth of the year.



Orbital launch attempts in 2026 by provider or country. Navy: success. Red: failure or partial failure.

CHART: THE LAUNCH REPORT · GCAT (JONATHAN MCDOWELL) THROUGH OCT 15; LAUNCH LIBRARY 2 SINCE

The failures

According to GCAT, seven launches have failed this year: India's PSLV on January 12; China's Long March 3B on January 16, Ceres-2 on January 17, Tianlong-3 on April 3 and Long

March 7A on August 10; Japan's Kairos on March 5; and Blue Origin's New Glenn on April 19, recorded as a partial failure.

From the sea

Tuesday's Ceres-1S flight was among eight successful Chinese orbital launches from sea platforms counted so far this year, using Jielong-3, Yinli-1 and Ceres-1S rockets. China's launch total is up 48% from the same point in 2024 (NASASpaceflight).

October so far

Eight of this month's 13 orbital launches were SpaceX Falcons (Launch Library 2).

DATE, EDT	ROCKET	MISSION	SITE
Oct 1, 11:10 a.m.	Falcon 9	Crew-13	Cape Canaveral
Oct 1, 2:32 p.m.	Falcon 9	Transporter-18 rideshare	Vandenberg
Oct 1, 11:54 p.m.	Falcon Heavy	NROL-97	Kennedy
Oct 6, 11:25 p.m.	Nuri	NeonSat-2 to 6	Naro, South Korea
Oct 9, 3:27 p.m.	Long March 12	SatNet LEO Group 27	Wenchang, China
Oct 10, 3:39 a.m.	Falcon 9	SDA Tranche 1 Transport Layer A	Vandenberg
Oct 12, 8:28 p.m.	Falcon 9	Starlink 15-25	Vandenberg
Oct 13, 6:33 a.m.	Falcon 9	CRS-35	Cape Canaveral
Oct 14, 7:40 a.m.	Zhuque-2E	10 satellites	Jiuquan, China
Oct 15, 8:01 p.m.	Falcon 9	USSF-481	Vandenberg
Oct 19, 3:41 p.m.	**H3-24**	**MMX**	**Tanegashima, Japan**
Oct 20, 7:04 a.m.	**Falcon 9**	**Bandwagon-5 rideshare**	**Cape Canaveral**
Oct 20, 11:39 p.m.	**Ceres-1S**	**Xiguang-4 01**	**Sea, off Rizhao, China**

The week ahead

Seven orbital launches have specific dates between Thursday and October 29 in Launch Library 2: three SpaceX Falcon 9s, three Long Marches and an Electron. If all fly on schedule, October would reach 20 orbital launches, SpaceX 123 for the year and Rocket Lab 17.

Takeaways

1. H3 launched MMX to Mars Monday; it arrives in August 2027.
2. Bandwagon-5 and a Ceres-1S sea launch also succeeded, for 246 attempts this year.
3. Starlink 15-15 was scrubbed twice and tries again at 5:45 p.m. EDT Thursday.
4. Rocket Lab is reviewing pad telemetry before its next Electron.
5. China is readying its first reflight of a recovered orbital booster.

KEY DATES AHEAD (EDT)

Thu Oct 22	Starlink 15-15, window opens 5:45 p.m.; New Glenn cadence meeting, 6–8 p.m.
Fri Oct 23	Electron for HawkEye 360, 3 a.m.; Long March 6A, 6:15 a.m.; Long March 4B, 3:25 p.m.
Sat Oct 24	Long March 8A, 9:50 p.m.
Mon Oct 26	Starlink 15-16, 1 a.m.
Thu Oct 29	Starlink 15-18, 12:17 a.m.; Vulcan with Amazon Leo, NET
Fri Oct 30	Rassvet-3 17 reentry predicted; New Glenn comments close
NET Nov 8	Starship Flight 15

SOURCES

Schedules: Launch Library 2 (The Space Devs), Spaceflight Now, NASASpaceflight, FAA operations plan advisory. Agencies: NASA, JAXA, ESA, FAA, Federal Register. Tracking: CelesTrak, The Aerospace Corporation, GCAT (Jonathan McDowell). X posts: @SpaceflightNow, @NASASpaceflight, @RocketLab, @JAXA_en, @blueorigin, @FireflySpace, @esa. Photos: NASA (public domain). Maps: Natural Earth (public domain).

The Launch Report is compiled from official agency releases, regulatory filings, orbital tracking data, public posts on X and published reporting. Quotes are reproduced verbatim unless marked as translated or paraphrased. Launch times change often; check with the launch provider before making plans.